

# Goodness-of-Fit & ML Hypothesis Testing

Logit and Probit Models

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# Learning Goals

By the end of this lecture, you should be able to:

- **Derive** the maximum likelihood estimator (**MLE**) in a simple binomial model and interpret it as a sample proportion.
- **Explain why** the usual OLS  $R^2$  is not appropriate for logit/probit models.
- **Define and interpret McFadden's pseudo- $R^2$**  and relate it to the log-likelihood.
- **Evaluate model performance** using classification tables, accuracy, sensitivity, and specificity.
- **Formulate and interpret LR, Wald, and LM** tests in ML frameworks.
- **Implement** these concepts in **Stata** (logit/probit, pseudo- $R^2$ , classification, LR/Wald tests).

## Example: Probability of Passing the Exam

### Data Summary (by exam round):

| Exam Round | Students | Passed | $\hat{\pi}$     |
|------------|----------|--------|-----------------|
| June (1st) | 95       | 71     | $\approx 0.747$ |
| July       | 38       | 21     | $\approx 0.553$ |
| September  | 27       | 10     | $\approx 0.370$ |
| February   | 15       | 12     | $\approx 0.800$ |

### Natural Questions:

- Does exam difficulty differ across rounds?
- Are better-prepared students taking the June exam?
- How confident are we in each estimate  $\hat{\pi}$ ?

# Modeling: Bernoulli Distribution

## Setup:

- For each student  $i$ :  $Y_i \in \{0, 1\}$  (fail or pass)
- All students in an exam round have same passing probability:  $P(Y_i = 1) = \pi$
- Students are independent:  $(Y_1, \dots, Y_n)$  are i.i.d.  $\text{Bernoulli}(\pi)$

## Probability mass function for each observation:

$$P(Y_i = y_i | \pi) = \pi^{y_i} (1 - \pi)^{1-y_i}$$

where  $y_i = 1$  if student passes,  $y_i = 0$  if student fails.

# Likelihood Function

**Joint probability of observing the entire sample:**

$$L(\pi; y_1, \dots, y_n) = \prod_{i=1}^n \pi^{y_i} (1 - \pi)^{1-y_i}$$

**Log-Likelihood (easier to work with):**

$$\ell(\pi; y) = \log L(\pi; y) = \sum_{i=1}^n [y_i \log \pi + (1 - y_i) \log(1 - \pi)]$$

**Why log-likelihood?**

- Avoids numerical underflow (products of small probabilities)
- Sums are easier to optimize than products
- Derivatives (scores) are simpler

# Maximum Likelihood Estimator (MLE)

**Goal:** Find  $\hat{\pi}$  that maximizes  $\ell(\pi; y)$ .

**First-order condition (Score = 0):**

$$\frac{\partial \ell(\pi; y)}{\partial \pi} = \frac{\sum_{i=1}^n y_i}{\pi} - \frac{\sum_{i=1}^n (1 - y_i)}{1 - \pi} = 0$$

Let  $n_1 = \sum_{i=1}^n y_i$  (number of students who passed). Then:

$$\frac{n_1}{\pi} = \frac{n - n_1}{1 - \pi}$$

Solving:

$$\hat{\pi}_{\text{ML}} = \frac{n_1}{n} = \bar{y}$$

**Conclusion:** The MLE of  $\pi$  is just the sample proportion!

# Deriving the MLE: Analytical Solution (1)

## Why showing the derivation?

- In this simple model, we can solve for  $\hat{\pi}$  **analytically** (by hand).
- This illustrates the general principle: finding parameters where the gradient is zero.

## Step-by-Step Solution:

### 0. Given the log-likelihood:

$$\ell(\pi; y) = \sum_{i=1}^n [y_i \log \pi + (1 - y_i) \log(1 - \pi)]$$

### 1. Step 1: Take the first derivative (Score function):

$$s(\pi) = \frac{\partial \ell(\pi; y)}{\partial \pi} = \sum_{i=1}^n \left[ \frac{y_i}{\pi} - \frac{1 - y_i}{1 - \pi} \right]$$

## Deriving the MLE: Analytical Solution (2)

2. **Step 2: Rearrange using  $n_1 = \sum y_i$  and  $n_0 = n - n_1$ :**

$$s(\pi) = \frac{n_1}{\pi} - \frac{n_0}{1 - \pi}$$

3. **Step 3: Set  $s(\pi) = 0$  and solve for  $\hat{\pi}$ :**

$$\frac{n_1}{\pi} - \frac{n_0}{1 - \pi} = 0 \quad \Rightarrow \quad \frac{n_1}{\pi} = \frac{n_0}{1 - \pi} \quad \Rightarrow \quad n_1(1 - \pi) = n_0\pi \quad \Rightarrow$$

$$\Rightarrow \quad n_1 - n_1\pi = n_0\pi \quad \Rightarrow \quad n_1 = (n_0 + n_1)\pi \quad \Rightarrow \quad \pi = \frac{n_1}{n_0 + n_1} \quad \Rightarrow \quad \boxed{\hat{\pi} = \frac{n_1}{n}}$$

**Crucial Difference with Logit/Probit:** In Logit/Probit, the Score equations are non-linear and **cannot be solved by hand**. Software must use iterative approximation (Newton-Raphson) to find the maximum.

## When is this Maximum? (Second-Order Condition)

**Second derivative (Hessian):**

$$H(\pi) = \frac{\partial^2 \ell(\pi; y)}{\partial \pi^2} = -\frac{n_1}{\pi^2} - \frac{n_0}{(1-\pi)^2} < 0 \quad \text{always}$$

**Since  $H(\pi) < 0$  everywhere:**

- The log-likelihood is concave
- There is a unique maximum
- $\hat{\pi} = n_1/n$  is indeed the MLE

**Intuition:** The second derivative measures how “peaked” the log-likelihood is. Negative everywhere means we have a unique, stable maximum.

# Computing the Hessian: Analytical Solution (1)

**Goal:** Verify that  $H(\pi) < 0$  by computing the second derivative from scratch.

## Step-by-Step Solution:

### 0. Starting with the Score (first derivative):

$$s(\pi) = \frac{\partial \ell(\pi; y)}{\partial \pi} = \frac{n_1}{\pi} - \frac{n_0}{1 - \pi}$$

where  $n_1 = \sum y_i$  and  $n_0 = n - n_1$ .

### 1. Step 1: Take the derivative of each term

$$\begin{aligned} H(\pi) &= \frac{\partial s(\pi)}{\partial \pi} = \frac{\partial}{\partial \pi} \left( \frac{n_1}{\pi} \right) + \frac{\partial}{\partial \pi} \left( -\frac{n_0}{1 - \pi} \right) \\ &= -\frac{n_1}{\pi^2} + \frac{\partial}{\partial \pi} \left( \frac{n_0}{1 - \pi} \right) \end{aligned}$$

## Computing the Hessian: Analytical Solution (2)

### 2. Step 2: Apply the quotient rule to the second term

For  $\frac{n_0}{1-\pi}$ : the numerator is constant, so

$$\frac{\partial}{\partial \pi} \left( \frac{n_0}{1-\pi} \right) = n_0 \cdot \frac{\partial}{\partial \pi} (1-\pi)^{-1} = n_0 \cdot (-1)(1-\pi)^{-2} \cdot (-1) = \frac{n_0}{(1-\pi)^2}$$

### 3. Step 3: Combine Step 1 and Step 2

$$H(\pi) = -\frac{n_1}{\pi^2} - \frac{n_0}{(1-\pi)^2}$$

✓ **Result:** Since  $n_1 > 0$ ,  $n_0 > 0$ ,  $\pi \in (0, 1)$ :

$$H(\pi) = -\frac{n_1}{\pi^2} - \frac{n_0}{(1-\pi)^2} < 0 \quad \text{always}$$

**Conclusion:** The log-likelihood is strictly concave  $\Rightarrow$  unique maximum at  $\hat{\pi}$ .

## Example Calculation

**June exam:**  $n = 95$  students,  $n_1 = 71$  passed

$$\hat{\pi}_{\text{June}} = \frac{71}{95} \approx 0.747$$

**September exam:**  $n = 27$  students,  $n_1 = 10$  passed

$$\hat{\pi}_{\text{Sept}} = \frac{10}{27} \approx 0.370$$

### Interpretation:

- June exam: students pass with probability  $\approx 75\%$  (easier exam)
- September exam: students pass with probability  $\approx 37\%$  (harder exam)
- Difference is substantial: might indicate different student ability or exam difficulty

# Confidence Intervals from ML

**Standard error of  $\hat{\pi}$ :**

$$SE(\hat{\pi}) = \sqrt{\frac{\hat{\pi}(1 - \hat{\pi})}{n}}$$

**95% Confidence Interval:**

$$\hat{\pi} \pm 1.96 \cdot SE(\hat{\pi})$$

**Example (June exam):**

$$SE = \sqrt{\frac{0.747 \times 0.253}{95}} \approx 0.044$$

$$95\% \text{ CI} = [0.747 - 1.96(0.044), 0.747 + 1.96(0.044)] = [0.661, 0.833]$$

**Example (September exam, smaller  $n$ ):**

$$SE = \sqrt{\frac{0.370 \times 0.630}{27}} \approx 0.093$$

$$95\% \text{ CI} = [0.370 - 1.96(0.093), 0.370 + 1.96(0.093)] = [0.188, 0.552]$$

Notice: **Wider CI with smaller  $n$**  (more uncertainty).

## Connection to Logit/Probit

In the binomial model we just studied:

$$P(Y_i = 1) = \pi \quad (\text{constant for all students})$$

In **logit/probit models**:

$$P(Y_i = 1 \mid X_i) = F(X_i\beta)$$

where:

- $F(\cdot)$  is a CDF (logistic or normal)
- The probability *varies* with observable characteristics  $X_i$
- $\beta$  is the parameter vector (like  $\pi$  before, but multidimensional)

**Key similarities:**

- Log-likelihood:  $\ell(\beta) = \sum_{i=1}^n [y_i \log F(X_i\beta) + (1 - y_i) \log(1 - F(X_i\beta))]$
- MLE: maximizes  $\ell(\beta)$
- Inference: uses scores, Hessian, LR/Wald/LM tests

# From Binomial Model to Model Comparison (1)

## What we learned so far:

- We can estimate  $\pi$  (a single parameter) via maximum likelihood
- At the MLE:  $\hat{\pi} = \bar{y}$  (sample mean)
- The log-likelihood tells us how good the fit is

## Now in logit/probit:

- Instead of one  $\pi$ , we have many parameters  $\beta$
- The MLE  $\hat{\beta}$  maximizes  $\ell(\beta)$
- Different models have *different log-likelihoods*

## From Binomial Model to Model Comparison (2)

### Key insight for model comparison:

- **Constant-only model:** Only intercept (no regressors)  $\Rightarrow \ell_{\text{const}}$
- **Full model:** All regressors included  $\Rightarrow \ell_{\text{full}}$
- **Always:**  $\ell_{\text{full}} \geq \ell_{\text{const}}$  (adding variables can't make fit worse)
- The **difference** in log-likelihoods measures improvement

# The Problem: $R^2$ in Nonlinear Models

**OLS:**  $R^2 = 1 - \frac{\sum \hat{u}_i^2}{\sum (y_i - \bar{y})^2}$  works great

**Logit/Probit:**  $R^2$  is **inappropriate** because:

- 1 Dependent variable is binary  $\Rightarrow$  fixed variation
- 2 Predictions  $\hat{P}_i \in (0, 1)$  are probabilities, not 0 or 1
- 3 Residuals  $\hat{u}_i = y_i - \hat{P}_i$  are not identically distributed

**Solution:** Use alternatives:

- Pseudo- $R^2$  (based on likelihoods)
- Classification accuracy (based on predictions)
- ML hypothesis tests (LR, Wald, LM)

# Log-Likelihood for Binary Models

For observation  $i$ :

$$P(Y_i = y_i | \mathbf{X}_i) = F(\mathbf{X}_i \beta)^{y_i} \cdot [1 - F(\mathbf{X}_i \beta)]^{1-y_i}$$

Sum over all  $n$  observations:

$$\ell(\beta; \mathbf{y}) = \sum_{i=1}^n [y_i \ln F(\mathbf{X}_i \beta) + (1 - y_i) \ln(1 - F(\mathbf{X}_i \beta))] \quad (1)$$

**Key facts:**

- $\ell(\beta; \mathbf{y}) < 0$  always (log of probabilities)
- **Larger (less negative)** = better fit
- **Logit:**  $F(z) = \frac{e^z}{1+e^z} = \frac{1}{1+e^{-z}}$
- **Probit:**  $F(z) = \Phi(z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^z e^{-t^2/2} dt$  (std normal)

# McFadden's Pseudo- $R^2$

The most common measure:

$$R_{\text{McF}}^2 = 1 - \frac{\ell_{\text{full}}}{\ell_{\text{const}}} \quad (2)$$

where:

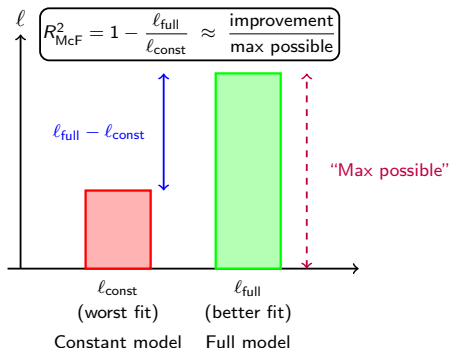
- $\ell_{\text{full}}$  = log-likelihood with all regressors
- $\ell_{\text{const}}$  = log-likelihood with only constant

## Interpretation:

- Ranges from 0 to 1 (approximately)
- $R_{\text{McF}}^2 = 0.15$ : 15% improvement over constant-only model
- **Benchmark:** 0.2–0.4 is “good” for discrete choice
- **NOT comparable to OLS  $R^2$**  (tends to be much lower)

# From Log-Likelihood Difference to McFadden's $R^2$

**Core idea:** Compare how much the log-likelihood improves when we move from the constant-only model to the full model.



## Interpretation:

- If the full model does not improve the log-likelihood  $\Rightarrow R^2_{\text{McF}} \approx 0$ .
- Larger improvements in log-likelihood  $\Rightarrow$  higher  $R^2_{\text{McF}}$ .

# Why McFadden's $R^2$ Makes Sense

Comparison with OLS  $R^2$ :

| Metric      | OLS                                       | Logit/Probit                                                            |
|-------------|-------------------------------------------|-------------------------------------------------------------------------|
| Numerator   | $\sum \hat{u}_i^2$                        | $\ell_{\text{full}}$                                                    |
| Denominator | $\sum (y_i - \bar{y})^2$                  | $\ell_{\text{const}}$                                                   |
| Intuition   | % of variance explained                   | % of likelihood improved                                                |
| Formula     | $R^2 = 1 - \frac{\text{SSR}}{\text{SST}}$ | $R^2_{\text{McF}} = 1 - \frac{\ell_{\text{full}}}{\ell_{\text{const}}}$ |

Why not use OLS  $R^2$  for logit/probit?

1. Binary  $y \in \{0, 1\}$  has **fixed variation** (no natural “total”)
2. Predictions  $\hat{P}_i \in (0, 1)$  are probabilities (not errors)
3. Residuals  $\hat{u}_i = y_i - \hat{P}_i$  are **heteroskedastic by design**
4. Log-likelihood is the natural scale for likelihood-based models

# McFadden's $R^2$ : Practical Guidelines

## Benchmark interpretation (from literature):

| Range     | Interpretation                                |
|-----------|-----------------------------------------------|
| 0.0 – 0.1 | Weak fit (consider different model)           |
| 0.1 – 0.2 | Fair fit (acceptable for exploratory work)    |
| 0.2 – 0.4 | <b>Good fit (typical for discrete choice)</b> |
| 0.4 – 0.6 | Very good fit (model does well)               |
| > 0.6     | Excellent fit (rare in practice)              |

## Important note:

- McFadden  $R^2$  is **NOT comparable to OLS  $R^2$**
- OLS  $R^2$  tends to be much *higher* than McFadden
- Example: OLS  $R^2 = 0.7 \approx$  McFadden  $R^2 = 0.15$  (same model!)
- Always report **McFadden + classification accuracy** together

## Alternative Pseudo- $R^2$ Measures

- **Cox-Snell:**  $R_{CS}^2 = 1 - \left( \frac{L_{\text{const}}}{L_{\text{full}}} \right)^{2/n}$  (uses likelihood, not log-likelihood)
- **Nagelkerke:**  $R_{\text{Nag}}^2 = \frac{R_{CS}^2}{R_{CS, \text{max}}^2}$  (scales to  $[0,1]$ )
- **Information Criteria:**
  - ▶  $\text{AIC} = 2k - 2\ell(\hat{\beta})$  (Smaller is better)
  - ▶  $\text{BIC} = k \ln(n) - 2\ell(\hat{\beta})$  (More penalty on  $k$ )

**In practice:** Report McFadden's  $R^2$  + classification accuracy

# Classification Table (Confusion Matrix)

## Process:

1. Compute predicted probability:  $\hat{P}_i = F(\mathbf{X}_i\hat{\beta})$
2. Apply threshold (typically 0.5):

$$\hat{y}_i = \begin{cases} 1 & \text{if } \hat{P}_i > 0.5 \\ 0 & \text{otherwise} \end{cases} \quad (3)$$

3. Tabulate against actual outcomes

|        |         | Predicted     |               |
|--------|---------|---------------|---------------|
|        |         | $\hat{y} = 0$ | $\hat{y} = 1$ |
| Actual | $y = 0$ | TN            | FP            |
|        | $y = 1$ | FN            | TP            |

# Accuracy Metrics

## Accuracy (Overall):

$$\text{Accuracy} = \frac{TP + TN}{n}$$

→ Proportion of correct predictions

## Sensitivity (True Positive Rate):

$$\text{Sensitivity} = \frac{TP}{TP + FN}$$

→ Proportion of actual 1's correctly predicted

## Specificity (True Negative Rate):

$$\text{Specificity} = \frac{TN}{TN + FP}$$

→ Proportion of actual 0's correctly predicted

**Trade-off:** Lower threshold  $\Rightarrow$  higher sensitivity, lower specificity

## Example: Ketchup Brand Choice (Model A)

Logit with pricebrand1 only

| brand1 | Predicted |   | Total |
|--------|-----------|---|-------|
|        | 0         | 1 |       |
| 0      | 2488      | 0 | 2488  |
| 1      | 310       | 0 | 310   |
| Total  | 2798      | 0 | 2798  |

**Analysis:**

- Accuracy:  $2488/2798 = 88.9\%$
- Sensitivity:  $0/310 = 0\%$  (predicts everyone as 0!)
- Specificity:  $2488/2488 = 100\%$

**Problem:** Model predicts everyone chooses  $y = 0$ . Weak predictive power.

## Example: Ketchup Brand Choice (Model B)

Logit with pricebrand1 AND pricebrand2

| brand1 | Predicted |    | Total |
|--------|-----------|----|-------|
|        | 0         | 1  |       |
| 0      | 2465      | 23 | 2488  |
| 1      | 309       | 1  | 310   |
| Total  | 2774      | 24 | 2798  |

Improved metrics:

- Accuracy:  $(2465 + 1)/2798 = 88.1\%$  (worse overall)
- Sensitivity:  $1/310 = 0.32\%$  (still very low)
- Specificity:  $2465/(2465 + 23) = 99.1\%$  (better)

**Insight:** Imbalanced data (mostly  $y = 0$ ) limits predictive ability

# The “Trinity” of ML Tests

Testing  $H_0 : \mathbf{R}\beta = \mathbf{q}$  vs  $H_1 : \mathbf{R}\beta \neq \mathbf{q}$

| Test        | Formula               | Needs         | When             |
|-------------|-----------------------|---------------|------------------|
| <b>LR</b>   | $-2(\ell_R - \ell_U)$ | Both          | Comparing models |
| <b>Wald</b> | (see next)            | Unrestr. only | Easy to estimate |
| <b>LM</b>   | (see next)            | Restr. only   | Rare in practice |

**Distribution:** All  $\sim \chi_q^2$  under  $H_0$ , where  $q = \#$  restrictions

**Decision:** If Test Stat  $> \chi_{q,\alpha}^2 \Rightarrow$  **Reject**  $H_0$

# Likelihood Ratio (LR) Test

## Formula:

$$LR = -2(\ell_R - \ell_U) \quad (4)$$

where  $\ell_R$  = restricted model,  $\ell_U$  = unrestricted model

## Intuition:

- If  $H_0$  true:  $\ell_R \approx \ell_U \Rightarrow LR \approx 0 \Rightarrow$  don't reject
- If  $H_0$  false:  $\ell_U \gg \ell_R \Rightarrow LR$  large  $\Rightarrow$  reject

## Advantages:

- Simple to compute (just log-likelihoods)
- Easy to interpret
- Standard for comparing nested models

## LR Test Example

**Setup:** Test  $H_0 : \beta_{\text{pricebrand2}} = 0$

- Restricted: logit with pricebrand1 only  $\Rightarrow \ell_R = -836.30363$
- Unrestricted: logit with pricebrand1 + pricebrand2  $\Rightarrow \ell_U = -715.62915$

**Calculation:**

$$\text{LR} = -2(-836.30363 - (-715.62915)) = -2 \times (-120.67448) = 241.35 \quad (5)$$

**Critical value** ( $q = 1, \alpha = 0.05$ ):  $\chi_{1,0.05}^2 = 3.841$

**Decision:**  $241.35 > 3.841 \Rightarrow \text{Reject } H_0$

**p-value:**  $\Pr(\chi_1^2 > 241.35) \approx 0.000$

# Wald Test

## Formula:

$$\text{Wald} = (\mathbf{R}\hat{\beta}_U - \mathbf{q})^\top [\mathbf{R}\widehat{\text{Var}}(\hat{\beta}_U)\mathbf{R}^\top]^{-1}(\mathbf{R}\hat{\beta}_U - \mathbf{q}) \quad (6)$$

→ Under  $H_0$ :  $\text{Wald} \sim \chi_q^2$

**In Stata** just use `test` command

```
logit brand1 pricebrand1 pricebrand2  
test pricebrand2 = 0
```

## Advantages:

- Only need unrestricted model
- Easiest to implement in practice
- Asymptotically equivalent to LR

# Lagrange Multiplier (LM) Test

## Formula:

$$LM = \mathbf{s}(\hat{\beta}_R)^\top [\mathbf{H}(\hat{\beta}_R)]^{-1} \mathbf{s}(\hat{\beta}_R) \quad (7)$$

where  $\mathbf{s}$  = score (gradient),  $\mathbf{H}$  = Hessian

## When to use:

- Restricted model is easy to estimate
- Unrestricted model is complex
- *Rarely used in practice* for binary choice

**Asymptotic equivalence:** LR, Wald, and LM all  $\sim \chi_q^2$  as  $n \rightarrow \infty$

## Practical Workflow: Step 1 - Load Data & Estimate Model

```
clear all  
use "ketchup_binary_clabe.dta"
```

```
* Baseline model  
logit brand1 pricebrand1  
est store model_A  
scalar ll_A = e(ll)
```

**Output:**  $\ell_A = -836.30363$

## Practical Workflow: Step 2 - Compute McFadden's $R^2$

\* Constant-only model

```
logit brand1
```

```
scalar ll_const = e(ll)
```

\* McFadden's R-squared

```
scalar mcfadden = 1 - (ll_A / ll_const)
```

```
display "McFadden's R-sq = " mcfadden
```

### Example:

- $\ell_{\text{const}} = -1086.1163$
- $\ell_A = -836.30363$
- $R_{\text{McF}}^2 = 1 - (-836.30363 / -1086.1163) = 0.230$

## Practical Workflow: Step 3 - Compute Predicted Probabilities

- \* Re-estimate model A

```
logit brand1 pricebrand1
```

- \* Predicted probability (logit formula)

```
gen phat_A = exp(_b[_cons] + _b[pricebrand1]*pricebrand1) / ///  
              (1 + exp(_b[_cons] + _b[pricebrand1]*pricebrand1))
```

- \* Classification (0.5 threshold)

```
gen yhat_A = (phat_A > 0.5)
```

- \* Classification table

```
tab brand1 yhat_A
```

## Practical Workflow: Step 4 - Extended Model & LR Test

\* Unrestricted model

```
logit brand1 pricebrand1 pricebrand2  
est store model_B  
scalar ll_B = e(ll)
```

\* LR test (manual)

```
scalar LR = -2 * (ll_A - ll_B)  
scalar df = 1  
scalar crit = invchi2(df, 0.95)  
scalar pval = chi2tail(df, LR)
```

```
display "LR = " LR  
display "Critical value = " crit  
display "P-value = " pval
```

## Practical Workflow: Step 5 - Wald Test

- \* Already estimated model B

```
logit brand1 pricebrand1 pricebrand2
```

- \* Single restriction

```
test pricebrand2 = 0
```

- \* Joint restrictions

```
test pricebrand1 pricebrand2
```

- \* Built-in LR test

```
lrtest model_A model_B
```

# Key Takeaways

## Goodness-of-Fit:

- ✓ McFadden's  $R^2 = 1 - \frac{\ell_{\text{full}}}{\ell_{\text{const}}}$  (benchmark: 0.2–0.4 is good)
- ✓ Classification accuracy (watch for imbalanced data)
- ✓ AIC/BIC for model comparison

## Hypothesis Testing:

- ✓ LR, Wald, LM all asymptotically  $\chi^2_q$
- ✓ LR: simple (just log-likelihoods)
- ✓ Wald: easiest in practice (Stata test command)
- ✓ All three asymptotically equivalent

## Practical Rule:

- ✓ Always report pseudo- $R^2$  + classification accuracy
- ✓ Manual computation = understanding
- ✓ Check imbalance in outcomes

# Exam Checklist

- ☐ McFadden's  $R^2$  formula
- ☐ Classification table (TN, TP, FN, FP)
- ☐ Accuracy, Sensitivity, Specificity
- ☐ LR test formula:  $LR = -2(\ell_R - \ell_U)$
- ☐ Distribution:  $\chi_q^2$
- ☐ Decision rule: compare to critical value
- ☐ Logit CDF:  $F(z) = \frac{e^z}{1+e^z} = \frac{1}{1+e^{-z}}$
- ☐ Probit CDF:  $F(z) = \Phi(z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^z e^{-t^2/2} dt$
- ☐ Manual computation in Stata using scalars
- ☐ Difference: LR vs Wald vs LM

# Questions?

*Let's apply this in Stata*